Cross-Era Generalization of Russian AI-Generated Text Detectors

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Abstract

Detectors of machine-generated Russian text are trained on corpora built with 2020–2021 generators, while the texts they meet today come from much newer models. We measure how such detectors transfer across corpora and generator eras. Encoders fine-tuned on CoAT, zero-shot Binoculars with Russian-capable model pairs, and encoders trained on corpus mixtures are evaluated on CoAT, LLMTrace, AINL-Eval 2025 and two new test sets generated with T-pro 2.0 and Qwen3.8-27B. ruRoBERTa-large trained on CoAT drops from 0.95 AUROC on CoAT to 0.52–0.64 on the newer corpora, and encoders trained on modern corpora score CoAT machine texts below human ones (0.46–0.48 AUROC). With Russian-capable model pairs, Binoculars separates free continuations from human text with up to 0.84 AUROC and paraphrases with at most 0.64; adding its features to a fine-tuned encoder changes AUROC by less than 0.04. Mixtures that include LLMTrace reach 0.79–0.82 AUROC on the 2026 test sets, but the training and test corpora share generator families, so the gain cannot be attributed to corpus diversity alone. Code: `https://github.com/egorsorvin/ru-ai-text-detection`.

1 Introduction

Detectors of AI-generated text are used in education, content moderation and fraud screening. For Russian, the main public resources come from the RuATD 2022 shared task and its extension CoAT [Shamardina et al., 2022, Shamardina et al., 2025], built with generators such as ruGPT-3 and ruT5. Generators have changed since then: current models write fluent text, and a detector that learned the artifacts of older models may miss it. In English, detection is known to degrade on unseen generators and domains [Dugan et al., 2024]; for Russian we found no evaluation across corpora of different eras.

We evaluate three kinds of detectors on three public corpora, CoAT, LLMTrace [Tolstykh et al., 2025] and AINL-Eval [Batura et al., 2025], under a leave-one-corpus-out protocol. We also build two test sets that pair CoAT human texts with outputs of 2026 generators, so the human side stays fixed and only the generator changes. Our contributions are:

- a cross-corpus and cross-era evaluation of Russian detectors, including two new held-out test sets;
- evidence that the shift goes both ways: ruRoBERTa-large trained on CoAT flags 14–17% of machine texts from 2026 generators, while encoders trained on modern corpora score ruGPT-3 outputs from CoAT as more human than human texts (0.32–0.38 AUROC);
- an analysis of Binoculars [Hans et al., 2024] for Russian: with Russian-capable pairs it separates free continuation from human text with 0.69–0.84 AUROC and paraphrase or simplification with 0.51–0.66, and a threshold fitted on CoAT gives 0.50–0.63 balanced accuracy on newer corpora;
- a negative result: adding Binoculars features to a multi-corpus encoder changes AUROC by -0.03 to $+0.03$ in a single run, and the gains from corpus mixtures are confounded with generator families shared between training and test corpora.

1.1 Team

Egor Sorvin designed the study, built the test sets, ran the experiments and wrote the report.

2 Related Work

Russian corpora. RuATD 2022 [Shamardina et al., 2022] and its extension CoAT [Shamardina et al., 2025] contain texts from 13 generators of 2020–2021 fine-tuned for translation, paraphrasing, summarization and simplification, plus free generation. LLMTrace [Tolstykh et al., 2025] covers generators of 2023–2025, including the GPT-4 family, GigaChat, YandexGPT and Qwen. AINL-Eval 2025 [Batura et al., 2025] contains scientific abstracts by GPT-4-Turbo, Gemma-2-27B, Llama-3.3-70B and withheld models. Each work evaluates detectors within its own corpus.

Generalization. M4 [Wang et al., 2024], MULTITuDE [Macko et al., 2023] and RAID [Dugan et al., 2024] show that trained detectors lose accuracy on unseen generators and domains. Kirichenko [Kirichenko, 2026] shows that on Russian data detector scores also follow the text source, so cross-corpus results mix a generator shift with a source shift.

Zero-shot detection. DetectGPT [Mitchell et al., 2023], Fast-DetectGPT [Bao et al., 2024] and Binoculars [Hans et al., 2024] use the probabilities of a language model instead of training. Binoculars detects over 90% of ChatGPT texts at a 0.01% false-positive rate in English, but LLMTrace reports an AI-class F1 of 0.02 on Russian. Paraphrasing evades such detectors [Krishna et al., 2023].

Published results are in Table 1. They are not directly comparable with ours: the official CoAT and AINL-Eval test labels are closed.

Work	Detector	Metric	Value
RuATD 2022, binary	TF-IDF + logistic regression	Accuracy	0.642
	BERT baseline	Accuracy	0.797
	Best system, fine-tuned encoders	Accuracy	0.830
	Human annotators	Accuracy	0.667
CoAT, binary	ruRoBERTa-large	Accuracy	0.864
	Human annotators	Accuracy	0.66
LLMTrace, Russian	Fine-tuned Mistral-7B	AI-class F1	0.986
	Binoculars, zero-shot	AI-class F1	0.02

Table 1: Published results on the corpora used in this work, as reported by their authors.

3 Model Description

Task. Given a Russian text s, a detector outputs a score $f(s)$; a higher score means the text is more likely machine-generated. Detectors are trained on the training split of one or more corpora and evaluated on the test split of every corpus. The question is how well f separates human and machine texts on corpora and generators that were not used for training.

We compare three detectors.

Fine-tuned encoder. The `[CLS]` vector h of a pretrained encoder (ruBERT-base or ruRoBERTa-large) goes through dropout and a linear layer to two logits. All weights are trained with cross-entropy.

Binoculars. For a text $s = x_1 \dots x_L$, an observer model O (base) and a performer model P (instruct version of the same family):

$$\ell_P(s) = -\frac{1}{L}\sum_i \log p_P(x_i \mid x_{<i}), \qquad \ell_X(s) = -\frac{1}{L}\sum_i \sum_v p_O(v \mid x_{<i}) \log p_P(v \mid x_{<i}),$$

$$B(s) = \ell_P(s)/\ell_X(s).$$

ℓ_P is the log-perplexity of the text, ℓ_X the expected log-perplexity of a typical text in the same contexts. A low B indicates machine text, so the detector score is $f(s) = -B(s)$. Nothing is trained. We use three pairs: Qwen3-4B, YandexGPT-5-Lite-8B and Qwen3-14B.

Encoder with Binoculars features. The vector $z = (B, \log \ell_P, \log \ell_X)$ from the Qwen3-14B pair is standardized on the training data, projected by a linear layer with GELU to 32 dimensions and concatenated with h before the classification layer. The encoder and both layers are trained; the Binoculars models stay frozen.

Training on corpus mixtures. The fine-tuned encoder and the encoder with Binoculars features are also trained on balanced samples of 16,000 texts per corpus from two or three corpora and tested on the corpus left out.

4 Dataset

Table 2 lists the data after our preprocessing.

Corpus	Train	Dev	Test	Machine share, test	Median words, test
CoAT	155,158	17,240	24,628	0.50	20
LLMTrace-ru	18,076	4,163	4,522	0.41	142
AINL-Eval	33,399	1,758	10,978	0.80	70
Gen: T-pro 2.0	–	–	5,642	0.53	23
Gen: Qwen3.8-27B	–	–	5,636	0.53	23

Table 2: Data statistics. Word counts are medians over all test texts.

Public corpora. CoAT (RussianNLP/coat, Apache-2.0): the official test labels are closed, so the HF validation split is our test and 10% of train, stratified by generator, is our dev. LLMTrace (iitolstykh/LLMTrace_detection): Russian part, official splits, texts labelled *mixed* removed. AINL-Eval (iis-research-team/AINL-Eval-2025, Apache-2.0): the official test labels are closed, so `dev_full` is our test and 5% of train is our dev; one empty text is removed.

Our test sets. We sampled 1,000 human texts of 15–200 words from the CoAT test split per task and generated one machine text per source with T-pro 2.0 (32B) and Qwen3.8-27B, reasoning mode off, temperature 0.7, top-p 0.9. The tasks are *continue* (the model continues the first 10 words of the human text), *paraphrase* and *simplify*. Outputs are cleaned by fixed rules: removal of chat preambles, markdown, reasoning blocks and repeated prompt words; rejection of refusals, non-Russian outputs, repetition loops and copies of the source. Continuations are cut at a sentence boundary to at most 1.2 times the source length. The rules rejected 6 of 3,000 T-pro outputs and 8 of 3,000 Qwen outputs. Each test set contains the machine texts and their human sources. Median length is 23 words for human texts and 21–32 words for machine texts, depending on the task. The sets are produced by `scripts/generate_set.py`; raw outputs are stored with the cleaned texts.

5 Experiments

5.1 Metrics

The main metric is AUROC, the probability that a random machine text gets a higher machine score than a random human text. It does not need a threshold and is not affected by class imbalance. We also report balanced accuracy, the mean of the true positive and true negative rates, and the true positive rate at a 5% false positive rate:

$$\mathrm{TPR@5\%} = \Pr\big(f(s) > t_{0.95} \mid s \text{ is machine}\big),$$

where $t_{0.95}$ is the 95th percentile of the scores f of human test texts.

5.2 Experiment Setup

Encoders are trained for 2 epochs with AdamW, weight decay 0.01, linear warmup over 6% of steps, gradient clipping at 1.0, bf16 and at most 256 tokens. The checkpoint with the best dev accuracy is evaluated once on every test set. ruBERT-base uses batch 32 and learning rate $2 \cdot 10^{-5}$; ruRoBERTa-large uses batch 16 and 10^{-5}. For corpus mixtures the dev set holds 1,500 balanced texts per training corpus and is evaluated every 300 steps. Trained detectors predict the class with the higher logit. Binoculars uses bf16 and at most 256 tokens; its accuracy threshold is fitted on CoAT dev. All results come from one run with seed 42. CoAT baselines were trained on an RTX 3060, the rest on one rented H100 80GB in about 3.5 hours of instance time. Code and exact commands are in the repository.

5.3 Baselines

TF-IDF over word 1–2-grams and character 2–5-grams, 300k features each, with logistic regression; ruBERT-base and ruRoBERTa-large fine-tuned on CoAT. Zero-shot Binoculars serves as the training-free baseline.

6 Results

CoAT. Table 3 compares our baselines with the published numbers. Our test split differs from the closed official one, and our TF-IDF adds character n-grams, so the comparison is approximate.

Transfer. Table 4 shows AUROC of all detectors on all test sets.

- **Old to new.** Detectors trained on CoAT reach 0.43–0.71 AUROC on the newer corpora and 0.55–0.62 on our sets. On our sets the human texts come from CoAT, so only the generator differs: ruRoBERTa-large keeps 93% of human texts correct but flags 14–17% of machine texts.

Detector	Accuracy, ours	Accuracy, published
TF-IDF + LR	0.759	0.642
ruBERT-base	0.859	0.797
ruRoBERTa-large	0.878	0.864

Table 3: Binary detection on CoAT. Published numbers: RuATD 2022 for TF-IDF and BERT, CoAT for ruRoBERTa.

Detector	CoAT	LLMTrace	AINL	Gen: Qwen3.8	Gen: T-pro
TF-IDF + LR, CoAT	0.85	0.43	0.71	0.57	0.55
ruBERT-base, CoAT	0.94	0.44	0.63	0.60	0.60
ruRoBERTa-large, CoAT	0.95	0.52	0.64	0.62	0.58
Binoculars Qwen3-4B	0.44	0.62	0.79	0.67	0.70
Binoculars YandexGPT-8B	0.40	0.62	0.80	0.57	0.60
Binoculars Qwen3-14B	0.43	0.61	0.78	0.68	0.70
CoAT + LLMTrace	0.92	0.99	**0.91**	0.81	0.80
+ Binoculars	0.92	0.99	**0.90**	0.81	0.79
CoAT + AINL	0.93	**0.67**	0.99	0.68	0.64
+ Binoculars	0.93	**0.68**	0.99	0.69	0.68
LLMTrace + AINL	**0.46**	0.99	0.99	0.82	0.81
+ Binoculars	**0.48**	0.99	0.99	0.80	0.81
All three	0.90	0.98	0.99	0.79	0.79
+ Binoculars	0.92	0.99	0.99	0.80	0.81

Table 4: AUROC. The last block is ruRoBERTa-large trained on corpus mixtures; bold marks the corpus left out of training. The generated sets are unseen by all detectors.

- **New to old.** Trained on LLMTrace and AINL, the encoder scores CoAT machine texts below human ones (0.46). By generator, ruGPT-3 outputs get 0.32–0.38 and translation systems 0.64–0.70. Binoculars shows the same pattern (0.40–0.44 overall). ruGPT-3 texts are about twice as long as the human texts, so length does not explain the inversion.
- **Mixtures.** Adding LLMTrace to CoAT raises AUROC on AINL from 0.64 to 0.91; adding AINL raises LLMTrace from 0.52 to 0.67. Mixtures with LLMTrace give the best AUROC on our sets, 0.82 on Qwen3.8 and 0.81 on T-pro, against at most 0.62 for CoAT-trained encoders and 0.70 for Binoculars. LLMTrace training data contains the GPT-4 family (37% of its machine texts), Llama-3.3, Gemma-2 and Qwen models, which also produced AINL and our sets, so these gains may come from covering the test generator families.
- **Binoculars features** change AUROC by -0.03 to $+0.03$, within run-to-run noise.

By task and length. Table 5 splits the results on our sets. Binoculars separates continuations but not paraphrases. The CoAT-trained encoder is weakest on continuations. For all detectors, texts of up to 20 words are the hardest; for the all-corpora encoder AUROC is 0.65–0.68 on them and 0.89–0.93 on texts longer than 40 words.

Detector	Continue	Paraphrase	Simplify	TPR@5%
ruRoBERTa-large, CoAT	0.58 / 0.52	0.62 / 0.60	0.66 / 0.62	0.14 / 0.11
Binoculars Qwen3-14B	0.77 / 0.84	0.63 / 0.63	0.66 / 0.62	0.14 / 0.21
All three corpora	0.84 / 0.85	0.74 / 0.71	0.79 / 0.82	0.38 / 0.36
+ Binoculars	0.86 / 0.86	0.76 / 0.75	0.78 / 0.83	0.38 / 0.39

Table 5: AUROC by task and TPR at 5% FPR on the generated sets, Qwen3.8 / T-pro.

Thresholds. With a threshold fitted on CoAT dev, Binoculars has 0.50–0.63 balanced accuracy on the newer corpora. A threshold fitted on the dev sets of the other modern corpora gives 0.57–0.72 on LLMTrace and AINL, within 0.04 of a threshold fitted on the test corpus itself. At a 5% false positive rate, the all-corpora encoder catches 92–95% of machine texts on LLMTrace and AINL, which it saw in training, and 36–38% on our sets.

Examples. Table 6 shows texts from the T-pro set with the machine probability from the CoAT-trained encoder and the all-corpora encoder.

Text	CoAT	All
Human. Его личность и работы вдохновляли поэтов и писателей, таких как Райнер Мария Рильке и Никос Казандзакис.	0.15	0.41
Continuation, first 10 words human. Март 2021- Сентябрь 2021 Два тренера, три зала, 6-8 тренировок в неделю. Силовые, кардио, растяжка – без перерывов. Пот, прогресс, первые результаты.	0.01	0.91
Paraphrase. Метеорологи обещают россиянам розовый апрель и персиковый лето! И потом, если что, не к чему будет придраться.	0.15	0.05

Table 6: Output samples: probability of the machine class.

7 Conclusion

We evaluated Russian AI-text detectors across three public corpora and two test sets generated with 2026 models. Detectors trained on 2020–2021 genera-

tors reach 0.43–0.71 AUROC on newer corpora and 0.55–0.62 on our sets, and detectors trained on modern corpora score the old corpus below chance, 0.46–0.48. Binoculars with Russian-capable pairs separates free continuations with up to 0.84 AUROC and paraphrases with at most 0.64, and its features change the AUROC of a fine-tuned encoder by at most 0.03. Training on corpus mixtures gives the best results on our test sets, 0.79–0.82 AUROC, but the corpora share generator families, and at a 5% false positive rate the best models catch under 45% of machine texts from the new generators. Open questions are a split that holds out whole generator families, several seeds, and generators outside the Qwen lineage.

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